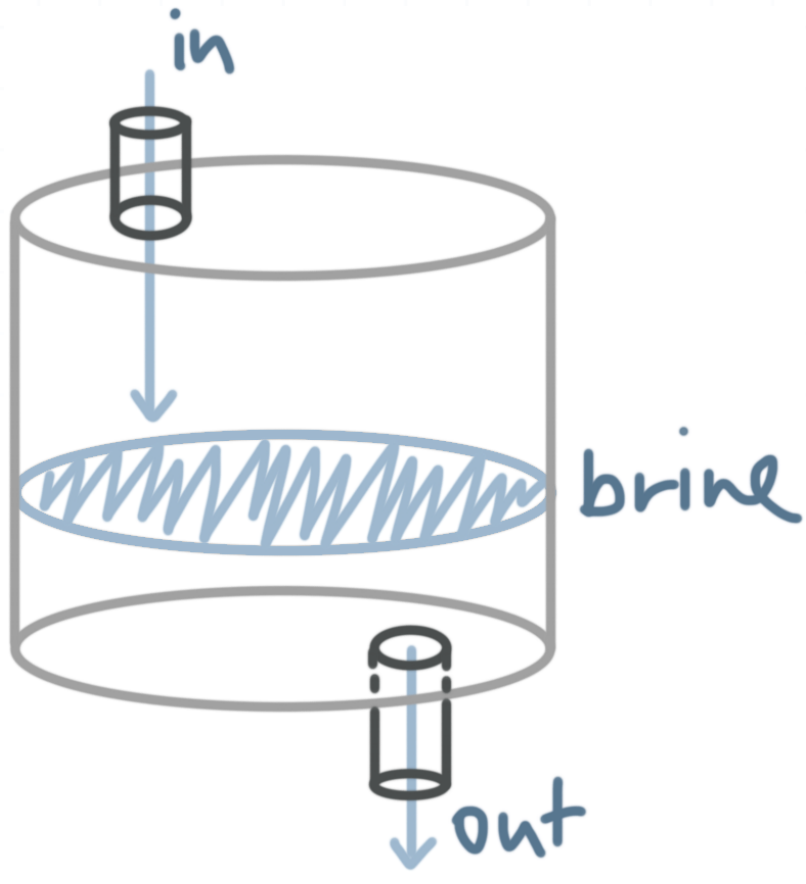
Mixing problems

Continuing our discussion about how we can use differential equations to model real-world questions, we want to look now at mixing problems, which can be modeled by separable equations.

The idea here is that we have some container holding some solution, like a tank holding a saltwater solution (a brine). We're feeding another saltwater solution into the tank through an input valve, while at the same time emptying the tank through an output valve.



Assuming the brine in the tank stays perfectly mixed at all times, we want to model this scenario with a differential equation, and then find the solution to the equation, which will be a function that describes the amount of salt in the tank at any given time.

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If salt is entering the tank at some rate, and exiting the tank at a different rate, then in general the amount of salt y at any time t can be modeled by

$$\frac{dy}{dt} = \text{(salt input rate)} - \text{(salt output rate)}$$

We can find the input rate of salt by multiplying the rate at which brine is entering the tank r_1 by the concentration of salt in that brine C_1.

$$\text{input rate} = C_1 r_1$$

As an example, if we're adding 5 gallons of brine to the tank every minute, $r_1 = 5$ gal/min, and the concentration of the input brine is $1/4$ pound of salt per gallon, $C_1 = 0.25$ lb/ gal, then the rate at which salt enters the tank is

$$\text{salt input rate} = \left(\frac{0.25 \text{ lb}}{\text{gal}}\right)\left(\frac{5 \text{ gal}}{\text{min}}\right) = \frac{1.25 \text{ lb}}{\text{min}}$$

Therefore, if C_1 is the concentration of the substance being added, r_1 is the rate at which that substance is added, if C_2 is the concentration of the substance being removed, and r_2 is the rate at which that substance is removed, then we can expand our salt concentration formula to

$$\frac{dy}{dt} = C_1 r_1 - C_2 r_2$$

It's important to realize that the concentration of salt leaving the tank is always modeled by

$$C_2 = \frac{y(t)}{\text{volume of brine in the tank at time } t}$$

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We know that $y(t)$ is the amount of salt at any specific time t, but the total volume in the tank can vary depending on the rate at which brine is being added to and emptied from the tank. But total salt divided by total brine volume will give the concentration C_2.

Input vs. output

In mixing problems like these ones, we'll handle three possible scenarios.

1. **brine input rate = brine output rate**: the total volume of brine in the tank at any time t is constant
2. **brine input rate > brine output rate**: the total volume of brine in the tank is increasing over time
3. **brine input rate < brine output rate**: the total volume of brine in the tank is decreasing over time

The differential equation we build to model each of these three cases will look slightly different, so let's do one example of each. Remember that to solve a separable equation, we'll separate variables, integrate the equation, and then solve for y to find the general solution.

Example

A tank contains $1{,}500$ L of water and 20 kg of dissolved salt. Fresh water is entering the tank at 15 L/min, and the solution drains at a rate of 10 L/min. Assuming the brine in the tank remains perfectly mixed, how much salt is in the tank after t minutes?

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Fresh water contains no salt at all, which means that $C_1 = 0$ kg/L because no salt is being added into the tank, but $r_1 = 15$ L/min because this is the rate at which fresh water is entering the tank.

And we know that $r_2 = 10$ L/min because this is the rate at which brine is being emptied from the tank.

To find C_2, we'll say that some amount of salt $y(t)$ is exiting the tank. But since C_2 is a concentration, we need to define the volume of the tank at any time t.

We start with $1{,}500$ L of brine solution in the tank. Then, every minute, 15 L enter the tank while 10 L exit. So every minute, we'll have 5 more liters of solution in the tank than we did the previous minute (this is the second of the three cases we outlined earlier, brine input rate > brine output rate).

Therefore, the volume of solution in the tank at any time is given by $1{,}500 + 5t$, and therefore the concentration of salt leaving the tank is

$$C_2 = \frac{y(t) \text{ kg}}{(1{,}500 + 5t) \text{ L}}$$

So the differential equation modeling the change in salt over time is

$$\frac{dy}{dt} = \left(\frac{0 \text{ kg}}{\text{L}}\right)\left(\frac{15 \text{ L}}{\text{min}}\right) - \left(\frac{y(t) \text{ kg}}{(1{,}500 + 5t) \text{ L}}\right)\left(\frac{10 \text{ L}}{\text{min}}\right)$$

$$\frac{dy}{dt} = \frac{0 \text{ kg}}{\text{min}} - \frac{10y(t) \text{ kg}}{(1{,}500 + 5t) \text{ min}}$$

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$$\frac{dy}{dt} = -\frac{2y(t)\ \text{kg}}{(300+t)\ \text{min}}$$

Now we'll separate variables.

$$dy = -\frac{2y}{300+t}\,dt$$

$$\frac{1}{y}\,dy = -\frac{2}{300+t}\,dt$$

With the variables separated, we'll integrate the equation.

$$\int \frac{1}{y}\,dy = \int -\frac{2}{300+t}\,dt$$

$$\ln|y| = -2\ln|300+t| + C$$

Simplify the equation to solve it for y.

$$e^{\ln|y|} = e^{-2\ln|300+t|+C}$$

$$|y| = Ce^{-2\ln|300+t|}$$

$$y = Ce^{\ln|300+t|^{-2}}$$

$$y = C|300+t|^{-2}$$

Then the general solution is

$$y(t) = \frac{C}{(300+t)^2}$$

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We were told that the tank initially contained 20 kg of dissolved salt, which means $y(0) = 20$ is an initial condition that we can substitute into the general solution to solve for the constant of integration,

$$20 = \frac{C}{(300+0)^2}$$

$$C = 1{,}800{,}000$$

and then we can put this value back into the general solution.

$$y(t) = \frac{1{,}800{,}000}{(300+t)^2}$$

This last example modeled the brine input rate > brine output rate case. Let's do the same example, but this time we'll switch the brine input and out rates, so that we can see what the result looks like in the brine input rate < brine output rate case.

Example

A tank contains $1{,}500$ L of water and 20 kg of dissolved salt. Fresh water is entering the tank at 10 L/min, and the solution drains at a rate of 15 L/min. Assuming the brine in the tank remains perfectly mixed, how much salt is in the tank after t minutes?

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In this scenario, the input rate is 10 L/min, while the output rate is 15 L/min. That means that the total volume in the tank is decreasing by 5 L every minute, which means the volume at any time t is given by $1{,}500 - 5t$.

$$\frac{dy}{dt} = \left(\frac{0 \text{ kg}}{\text{L}}\right)\left(\frac{10 \text{ L}}{\text{min}}\right) - \left(\frac{y(t) \text{ kg}}{(1{,}500 - 5t) \text{ L}}\right)\left(\frac{15 \text{ L}}{\text{min}}\right)$$

$$\frac{dy}{dt} = \frac{0 \text{ kg}}{\text{min}} - \frac{15y(t) \text{ kg}}{(1{,}500 - 5t) \text{ min}}$$

$$\frac{dy}{dt} = -\frac{3y(t) \text{ kg}}{(300 - t) \text{ min}}$$

Separate variables and integrate both sides,

$$\int \frac{1}{y}\, dy = \int -\frac{3}{300 - t}\, dt$$

$$\ln|y| = 3\ln|300 - t| + C$$

then solve for y.

$$e^{\ln|y|} = e^{3\ln|300-t|+C}$$

$$|y| = Ce^{3\ln|300-t|}$$

$$y = Ce^{\ln|300-t|^3}$$

$$y = C|300 - t|^3$$

Then the general solution is

$$y(t) = C|300 - t|^3$$

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We were told that the tank initially contained 20 kg of dissolved salt, which means $y(0) = 20$ is an initial condition that we can substitute into the general solution to solve for the constant of integration,

$$20 = C|300 - 0|^3$$

$$C = \frac{1}{1{,}350{,}000}$$

and then we can put this value back into the general solution.

$$y(t) = \frac{|300 - t|^3}{1{,}350{,}000}$$

Finally, let's look at the easiest example, the brine input rate = brine output rate case.

Example

A tank contains $1{,}500$ L of water and 20 kg of dissolved salt. Fresh water is entering the tank at 15 L/min, and the solution drains at a rate of 15 L/min. Assuming the brine in the tank remains perfectly mixed, how much salt is in the tank after t minutes?

In this scenario, the input rate is 15 L/min, while the output rate is 15 L/min. That means that the total volume in the tank remains the same, which means the volume at any time t is given by $1{,}500$.

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$$\frac{dy}{dt} = \left(\frac{0 \text{ kg}}{\text{L}}\right)\left(\frac{15 \text{ L}}{\text{min}}\right) - \left(\frac{y(t) \text{ kg}}{1{,}500 \text{ L}}\right)\left(\frac{15 \text{ L}}{\text{min}}\right)$$

$$\frac{dy}{dt} = \frac{15 \text{ L}}{\text{min}}\left(\frac{0 \text{ kg}}{\text{L}} - \frac{y(t) \text{ kg}}{1{,}500 \text{ L}}\right)$$

$$\frac{dy}{dt} = -\frac{y(t) \text{ kg}}{100 \text{ min}}$$

Separate variables and integrate both sides,

$$\int \frac{1}{y}\, dy = \int -\frac{1}{100}\, dt$$

$$\ln|y| = -\frac{1}{100}t + C$$

then solve for y.

$$e^{\ln|y|} = e^{-\frac{1}{100}t+C}$$

$$|y| = Ce^{-\frac{1}{100}t}$$

Then the general solution is

$$y(t) = Ce^{-\frac{1}{100}t}$$

We were told that the tank initially contained 20 kg of dissolved salt, which means $y(0) = 20$ is an initial condition that we can substitute into the general solution to solve for the constant of integration,

$$20 = Ce^{-\frac{1}{100}(0)}$$

$$C = 20$$

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and then we can put this value back into the general solution.

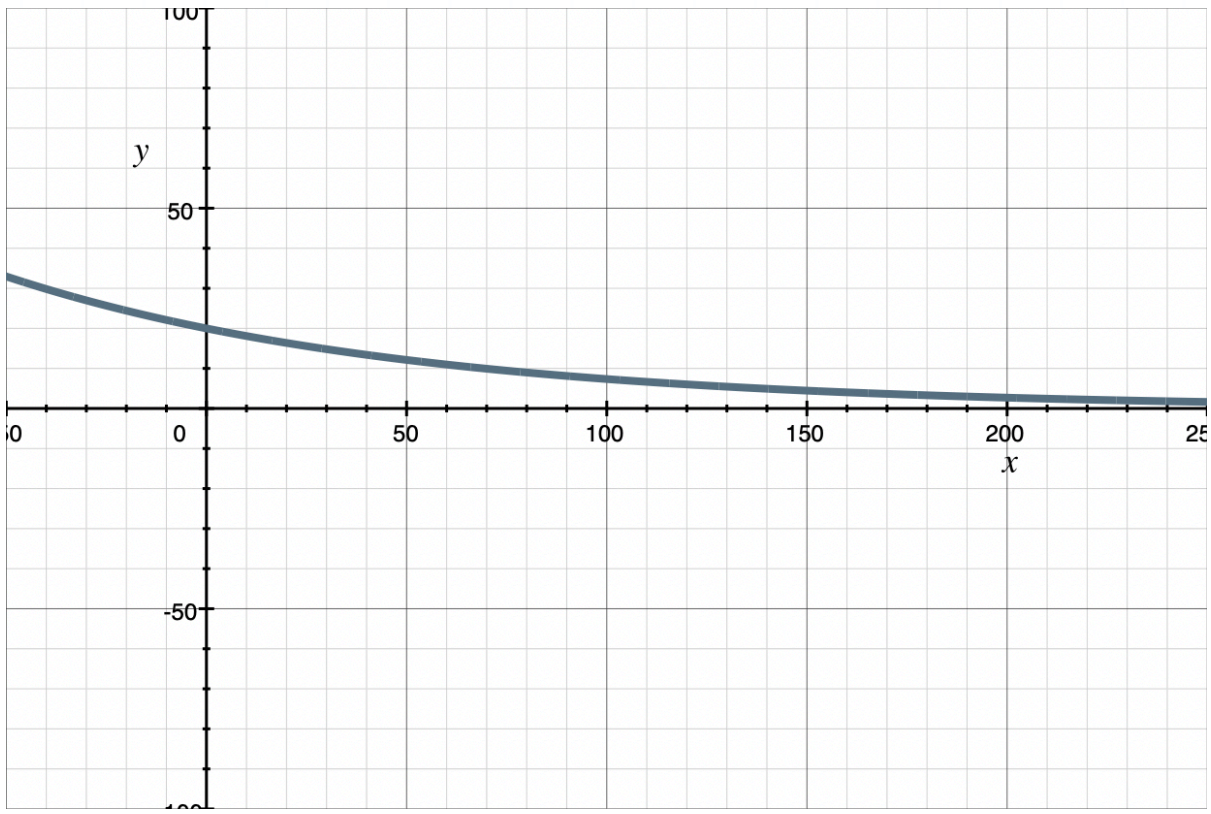
$$y(t) = 20e^{-\frac{1}{100}t}$$

If we graph the three solution equations from these examples, we can see how the amount of salt changes in the tank over time.

Notice how all three curves intersect the y-axis at $(0,20)$, matching the initial condition that we used in each problem. Based on the graphs, here's a summary of how the amount of salt changes over time in each case:

brine input rate = brine output rate: salt is initially 20 kg, but decreases over time

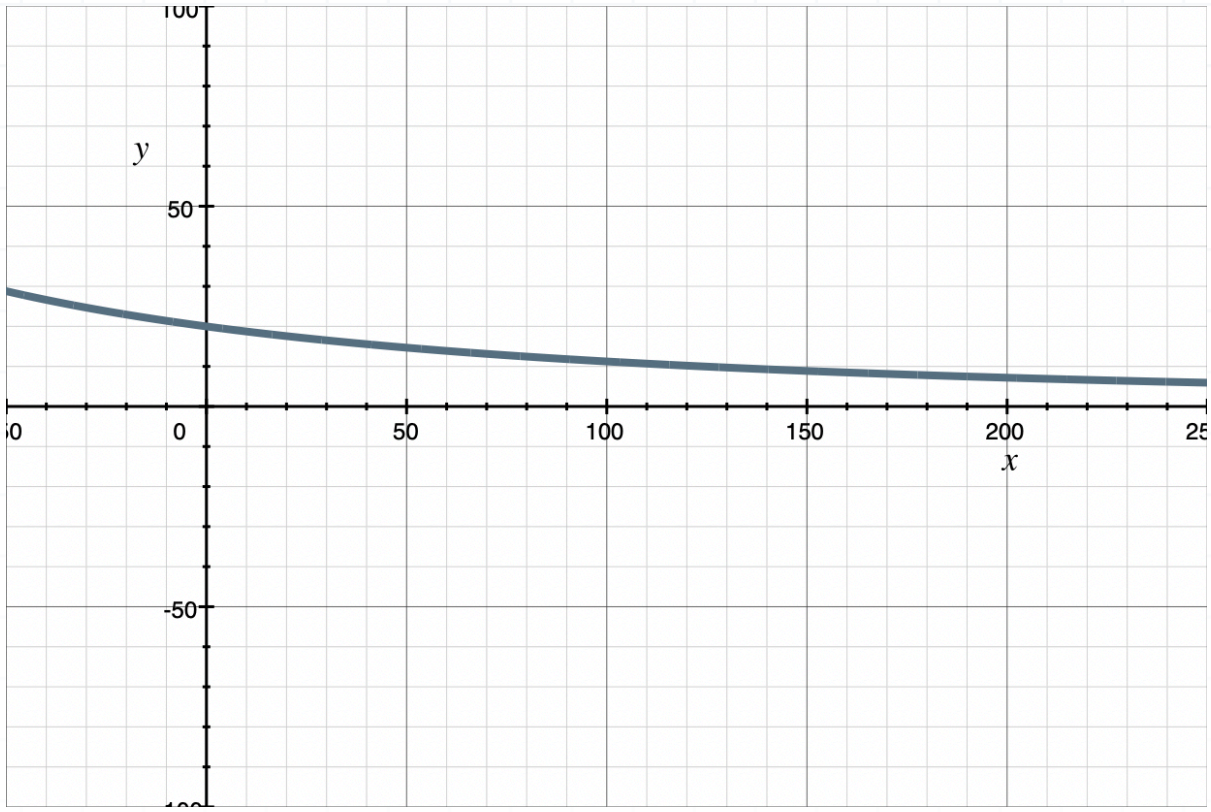
$$y(t) = 20e^{-\frac{1}{100}t}$$



brine input rate > brine output rate: salt is initially 20 kg, but decreases over time

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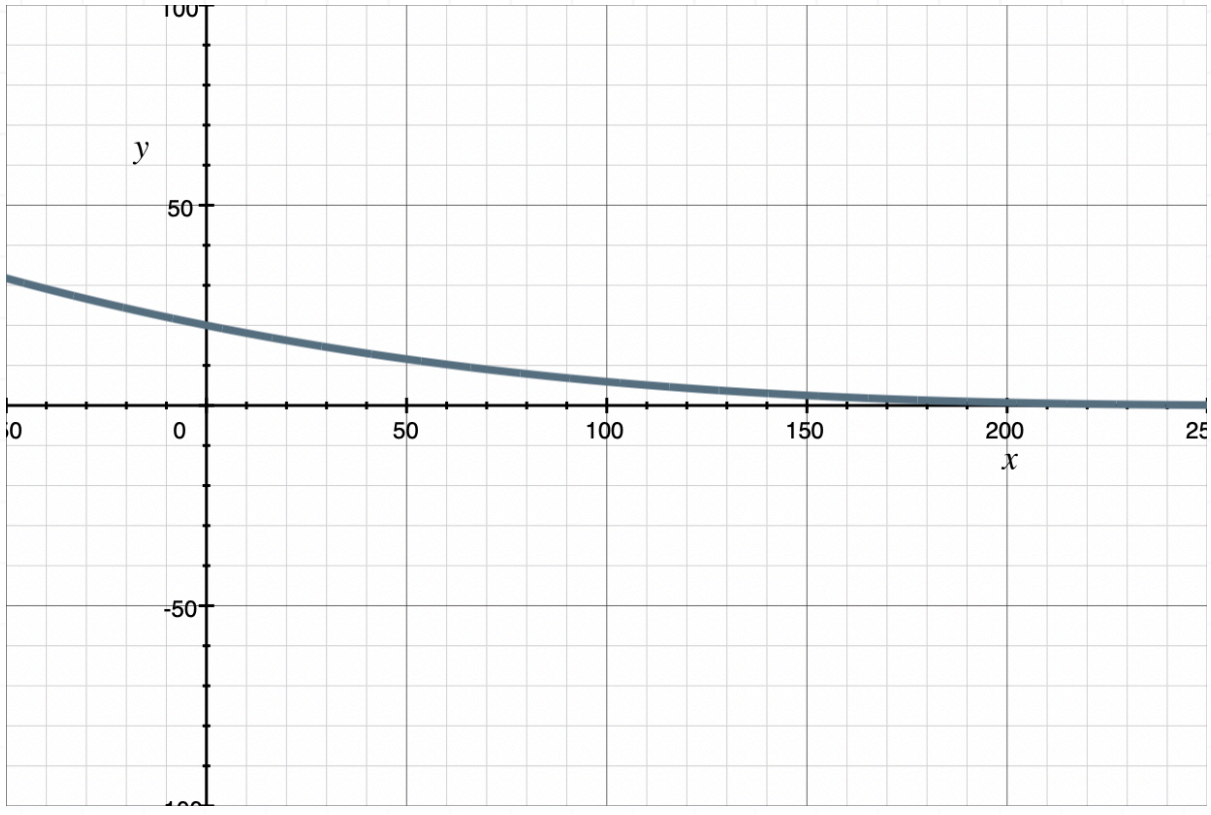
$$y(t) = \frac{1{,}800{,}000}{(300+t)^2}$$



brine input rate < brine output rate: salt is initially 20 kg, but decrease over time

$$y(t) = \frac{|300-t|^3}{1{,}350{,}000}$$

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